Universal gravitation

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Predict how gravitational force changes with mass and separation.

SUCCESS CRITERIA - I can...

1. Use center-to-center separation.
2. Apply direct mass and inverse-square distance comparisons.
3. Identify equal forces on the two interacting bodies.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “gravity” and “separation” to explain your answer. Sentence starters are optional.

gravity: an attractive interaction between masses

separation: distance between the specified objects or centers

inverse square: proportional to one divided by distance squared

01 Warm-up: bring back an earlier idea

A rover travels around a bend at a steady **5 m/s**. **Describe an observation** that shows velocity changes even though the speed reading does not.

Sentence starter: I notice … . This supports … because … .

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $F = Gm_1m_2/r^2$.
r is center-to-center separation. At fixed masses, doubling r makes the attractive force one quarter as large.

Worked reasoning: If separation doubles while both masses stay fixed, gravitational force becomes $\mathbf{1/2^2=1/4}$ as large. A **20 N** interaction would become **5 N**.

Your turn: 1. **Sketch the two separations** using centers.
2. **Draw equal** and opposite force arrows for each pair.

Sentence starter: In my work, I used … to show … . I labeled it … .

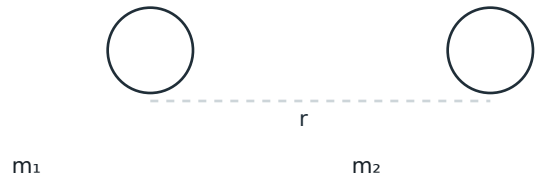


Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

Two equal spheres are shown. **Mark the distance r used** by the gravity equation and **draw a force** on each sphere.

Sentence starter: In my work, I used … to show … . I labeled it … .



Each circle represents one of the labeled objects or charges; r is their center-to-center separation.

Marzano check - circle: ① ② ③

04 Apply the physics

A gravity simulator shows a **12 N** attraction. Run two separate comparisons from that starting setup: (a) double one mass; (b) double the center-to-center distance. Keep everything else fixed in each trial. **Complete the force column**.

Sentence starter: I notice … . This supports … because … .

Trial	Change from start	Force (N)
Start	None	12
A	One mass ×2	____
B	Distance ×2	____

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Explain your reasoning

Does a large planet exert more force on a small moon than the moon exerts on the planet? **Compare forces** and accelerations.

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A runner pushes the ground backward with **60 N**. **Name the partner force**, its size and the object it acts on. **Explain its role** in the runner's motion.
Put each partner force on the object it acts on.

Sentence starter: I think … . I can point to …, which shows …, so … .



Object A



Object B

Marzano check - circle: 1 2 3

07 Exit check

At separation r the force is **8 N**. At **2** r with masses unchanged, what is it?

Sentence starter: I notice … . This supports … because … .





Mark the new separation.

Mass 1

Mass 2

Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

Starting from a force of **12 N**, double one mass and double the separation. **Find the new force** and **explain both effects separately**.

Sentence starter: I think … . I can point to …, which shows …, so … .

Quantity	Original	New
m_1	m_1	____
m_2	m_2	____
r	r	____
Force	F_0	____

Marzano check - circle: 1 2 3